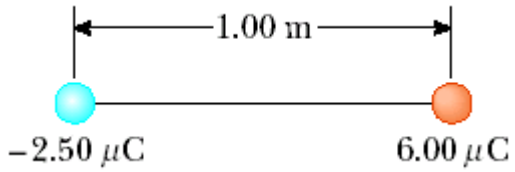
Physics 210 Tutorial 2

These problems are from an old text. The numbers match the numbers in the solutions.

15. In the figure below, determine the point (other than infinity) at which the electric field is zero.



16. An airplane is flying through a thundercloud at a height of 2 000 m. (This is a very dangerous thing to do because of updrafts, turbulence, and the possibility of electric discharge.) If a charge concentration of +40.0 C is above the plane at a height of 3 000 m within the cloud and a charge concentration of –40.0 C is at height 1 000 m, what is the electric field at the aircraft?

25. A rod 14.0 cm long is uniformly charged and has a total charge of –22.0 μC. Determine the magnitude and direction of the electric field along the axis of the rod at a point 36.0 cm from its center.

26. A continuous line of charge lies along the x axis, extending from $x = +x_0$ to positive infinity. The line carries charge with a uniform linear charge density λ_0. What are the magnitude and direction of the electric field at the origin? Hint: Just change the limits of integration to $+x_0$ to infinity.

27. A uniformly charged ring of radius 10.0 cm has a total charge of 75.0 μC. Find the electric field on the axis of the ring at (a) 1.00 cm, (b) 5.00 cm, (c) 30.0 cm, and (d) 100 cm from the center of the ring.

28. A line of charge starts at $x = +x_0$ and extends to positive infinity. The linear charge density is $\lambda = \lambda_0 x_0 / x$. Determine the electric field at the origin.

29. Show that the maximum magnitude E_{max} of the electric field along the axis of a uniformly charged ring occurs at $x = a/\sqrt{2}$ and has the value $Q/\left(6\sqrt{3}\pi\varepsilon_0 a^2\right)$. Where a is the radius of the ring and Q is the total charge of the ring. Hint: Take the derivative of E with respect to x and set it to zero, then solve for x and put that value back into the formula for E.

30. A uniformly charged disk of radius 35.0 cm carries charge with a density of 7.90×10^{-3} C/m^2. Calculate the electric field on the axis of the disk at (a) 5.00 cm, (b) 10.0 cm, (c) 50.0 cm, and (d) 200 cm from the center of the disk.

32. Show that the electric field of a uniformly charged disk of radius R and total charge Q at distances x that are large compared with R approaches that of a point charge $Q = \sigma\pi R^2$. (*Suggestion:* First show that $x / (x^2 + R^2)^{1/2} = (1 + R^2 / x^2)^{-1/2}$ and use the binomial expansion $(1 + \delta)^n \approx 1 + n\delta$ when $\delta << 1$.)

33. A uniformly charged insulating rod of length 14.0 cm is bent into the shape of a semicircle as shown below. The rod has a total charge of –7.50 μC. Find the magnitude and direction of the electric field at O, the center of the semicircle.

